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PROFESSIONAL SUMMARY

MS Computer Science graduate and AI/ML Engineer with 4+ years of hands-on experience building production LLM applications, RAG pipelines, agentic AI systems, and ML model deployment workflows across financial services and enterprise support domains. Proficient in Python, PyTorch, Hugging Face Transformers, LangChain, and vector databases (ChromaDB, Pinecone); experienced fine-tuning BERT-family models with LoRA/PEFT and deploying real-time inference APIs on AWS and Azure. Interested in Responsible AI, LLM safety evaluation, and publishing research insights at the intersection of data quality and frontier model development

CORE SKILLS

LLMs & NLP: Transformer Fine-Tuning (BERT, RoBERTa), LoRA, PEFT, RAG Pipelines, LangChain, LangGraph, Agentic AI, NER, Sentiment Analysis, Intent Classification, spaCy, Hugging Face
ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, CNNs, LSTMs, Optuna, Bayesian Optimization, MLflow, SHAP
Vector DBs & Retrieval: ChromaDB, FAISS, Pinecone, BM25, SentenceTransformers, Hybrid Retrieval, Embedding Pipelines
Data Engineering: Python, Pandas, NumPy, PySpark, Apache Kafka, Airflow, PostgreSQL, Supabase, SQLAlchemy, AWS Glue
Cloud & MLOps: AWS (SageMaker, Lambda, EC2, ECS, RDS), Azure ML, Docker, Kubernetes, GitLab CI/CD, GitHub Actions, Prometheus, Grafana
Backend & APIs: FastAPI, Flask, Django, REST APIs, Pydantic, Node.js, OCR (Mindee, Google Vision)
Frontend & Visualization: React, TypeScript, Next.js, Streamlit, Plotly, Figma

PROFESSIONAL EXPERIENCE

Applied AI Engineer

Feb 2024 – Present

MultiEdge, Inc. (Client: Centific)

Remote

- Contributed to Citizens Bank’s **AI-driven Customer Experience Platform**, enhancing personalized banking and financial advisory through NLP and recommendation systems; collaborated with cross-functional teams in requirement gathering sessions to align AI features with business objectives.
- Designed and deployed **scalable data ingestion pipelines** using Python (BeautifulSoup, Pydantic), SQLAlchemy, and AWS Glue, processing millions of customer interaction logs, transaction histories, and product usage records into PostgreSQL with complete data lineage for SOC 2 and PCI DSS audit readiness.
- Engineered behavioral features including **intent classification, sentiment scoring, and transaction pattern analysis** using pandas, spaCy, and custom rule-based extractors, improving downstream model signal quality by 27%.
- Developed and fine-tuned **transformer-based NLP models** (BERT, RoBERTa) on AWS SageMaker for intent detection, query classification, and personalized financial product recommendations, increasing model accuracy from 0.78 to 0.90 through domain-specific tuning.
- Applied **LoRA and PEFT** via Hugging Face Accelerate on AWS EC2, enabling efficient fine-tuning on proprietary banking datasets while maintaining stringent data privacy and governance standards.
- Executed 100+ **hyperparameter optimization experiments** using Optuna (Bayesian search, early stopping, k-fold CV), tracking results via MLflow and integrating with GitLab CI/CD for seamless reproducible model deployment.
- Implemented **real-time conversational AI pipelines** using LangChain, FAISS, and AWS Lambda to power intelligent banking chatbots for account inquiries and financial guidance, driving a 22% increase in customer satisfaction scores.
- Built **interactive analytics dashboards** using Streamlit and Plotly on AWS Elastic Beanstalk, enabling product and customer success teams to visualize chatbot engagement metrics and AI-driven insights, reducing manual escalations by 33%.

AI/ML Engineer

Feb 2022 – Jun 2023

EPAM Systems (Client: Dell Technologies)

Hyderabad, India

- Led development of the **Customer Support Intelligence System** an ML-powered solution automating ticket categorization, sentiment analysis, and response prioritization for Dell’s global support teams, reducing average response time by 20% and ticket backlog by 15%.

- Built **ETL pipelines** processing customer support tickets, chat logs, and product usage feedback using Python, PySpark, and Azure Data Factory, producing clean labeled datasets for model training across 3 Dell product lines.
- Developed and deployed **NLP classification models** (scikit-learn, XGBoost, BERT) to auto-categorize tickets by issue type and predict urgency from sentiment, achieving F1-score of 0.87 for ticket classification and 90% accuracy in sentiment detection.
- Applied **Grid Search and SHAP explainability** to tune model hyperparameters and ensure transparent, interpretable predictions — a key requirement for Dell's support compliance and QA review process.
- Deployed models via **Azure ML** with Docker and Kubernetes, integrating real-time scoring REST APIs into Dell's ticket creation platform; configured **Apache Airflow** for monthly model retraining with Prometheus/Grafana performance tracking.
- Partnered with Dell's Support Operations and Customer Experience teams to validate results and execute a phased rollout, resulting in measurable improvements in customer satisfaction scores and agent productivity.

Graduate Teaching Assistant – Software Engineering

Aug 2023 – Dec 2023

University of North Texas

Denton, TX

- Mentored 30+ graduate students on ML system design, Python, and REST API development; reviewed code submissions and provided structured feedback on architecture and documentation standards.

PERSONAL PROJECTS

DriveAssist AI – Production RAG Application with Hybrid Retrieval

2025–2026

- *Tech: Python, LangChain, ChromaDB, BM25, OpenAI GPT-4o, FastAPI, React 19, SentenceTransformers* | [Link](#)
- Built a **full-stack RAG system** enabling natural language Q&A over 600-page vehicle owner's manuals using a 7-layer pipeline: PDF ingest → chunking (700-char, RecursiveTextSplitter) → local SentenceTransformer embeddings → ChromaDB vector store → hybrid BM25+semantic retrieval → GPT-4o generation (temperature=0) → FastAPI + React chat UI.
- Designed a **hybrid retrieval ensemble** (50/50 BM25 + ChromaDB) that outperforms either method alone: BM25 catches exact keyword matches (part numbers, error codes) while semantic search handles meaning-based queries; used local embeddings to eliminate API embedding costs regardless of document size.
- Shipped a **dark automotive-themed React 19 + Vite frontend** with cited, grounded answers and page references; retriever indexes cached per-session to avoid rebuild overhead on every chat message.

ResumeAI – Full-Stack AI Resume Optimization Platform

2025–Present

- *Tech: React, TypeScript, FastAPI, Supabase, Redis, Claude API, OpenAI API, Python* | [Link](#)
- Built an **AI-powered resume platform** where users upload PDF/DOCX resumes, input a job description, and receive ATS-aligned, bullet-level rewrite suggestions via Claude/OpenAI APIs with real-time streaming progress.
- Implemented an **accept/reject/edit review workflow** with tailored PDF/DOCX export, resume history tracking, and AI-generated cover letters; backend uses Redis for session caching and Supabase for persistent user data.

Proofi AI – Credential Management & Showcase Platform (Beta)

2025–Present

- *Tech: Next.js 14, TypeScript, Tailwind CSS, Prisma, Supabase, NextAuth.js, Google Gemini, Vercel* | [Link](#)
- Co-founded and shipped a **live SaaS platform** where professionals claim a custom URL to showcase certifications; Google Gemini AI auto-extracts issuer, date, and credential ID from uploaded certificate images — zero manual typing.
- Built **AI-powered credibility verification** that confirms certificate authenticity before publishing; includes learning streaks, annual goal tracking, and personalized upskilling recommendations for registered users.

B0Bot – Agentic Cybersecurity Intelligence Platform (Open Source)

2024–Present

- *Tech: Python, Flask, Kafka, LangChain, LangGraph, Pinecone, Redis, Figma* | [Link](#)
- Open-source contributor on the **c2siorg b0bot project**: designed full application UI/UX in Figma, defined system architecture, authored project documentation, and built the agentic LangGraph pipeline for autonomous multi-source threat intelligence aggregation.
- Engineered a **real-time CDC pipeline** (Kafka + Debezium) for ingestion and deduplication, Pinecone vector storage with Redis caching, and LangGraph agents for automated threat classification and email digest delivery.

ACADEMIC PROJECTS

Predicting Rainfall in Australia Using Machine Learning

May – Jul 2024

- Developed and benchmarked 6 classifiers (Decision Trees, SVM, Naïve Bayes, KNN, XGBoost, Random Forest) on a multi-station Australian weather dataset; used pandas, scikit-learn, and matplotlib for preprocessing, EDA, and evaluation across accuracy, precision, recall, and F1 metrics to identify the strongest model for next-day rainfall prediction.

Predictors of Movie Box Office Success: Empirical Analysis

Jan – May 2024

- Analyzed revenue drivers across 5,000+ films using Multiple Linear Regression, Random Forests, and Gradient Boosting with pandas, NumPy, and scikit-learn; identified budget and social media metrics as dominant success predictors, delivering actionable insights for studio investment decisions.

Weather Impact on Uber/Lyft Ride Demand

Sep 2023

- Applied KNN, Logistic Regression, and SVM to Boston ride-share and weather datasets, achieving 87% predictive accuracy; surfaced actionable patterns between weather conditions and ride-type preferences to inform dynamic pricing strategy.

AssistMap – Accessible Restroom Locator Web Application

Aug – Dec 2023

- Built a full-stack accessibility web app using Django, PostgreSQL, AWS EC2/RDS/SES, and JavaScript with real-time geolocation search, Maps API integration, secure user auth, and production Unicorn/Nginx deployment on AWS EC2. | [Link](#)

EDUCATION

University of North Texas

Denton, TX

Master of Science in Computer Science

Aug 2023 – Dec 2024

CERTIFICATIONS

AWS Certified AI Cloud Practitioner – Amazon Web Services

AWS Certified Developer – Associate – Amazon Web Services

Google AI Professional Certification – Google

Docker Professional Certificate – Docker